Erasure conversion for quantum fault tolerance

Metastable alkaline-earth(-like) atoms convert dominant gate faults into known-location erasures

One-line thesis If hardware can **flag where faults happened**, surface-code decoding becomes much easier and thresholds rise dramatically.

Erasure fraction

$R_e \approx 0.98$

estimated in proposal

Surface-code threshold

0.937% → 4.15%

proposal summary

Core mechanism

bright-vs-dark

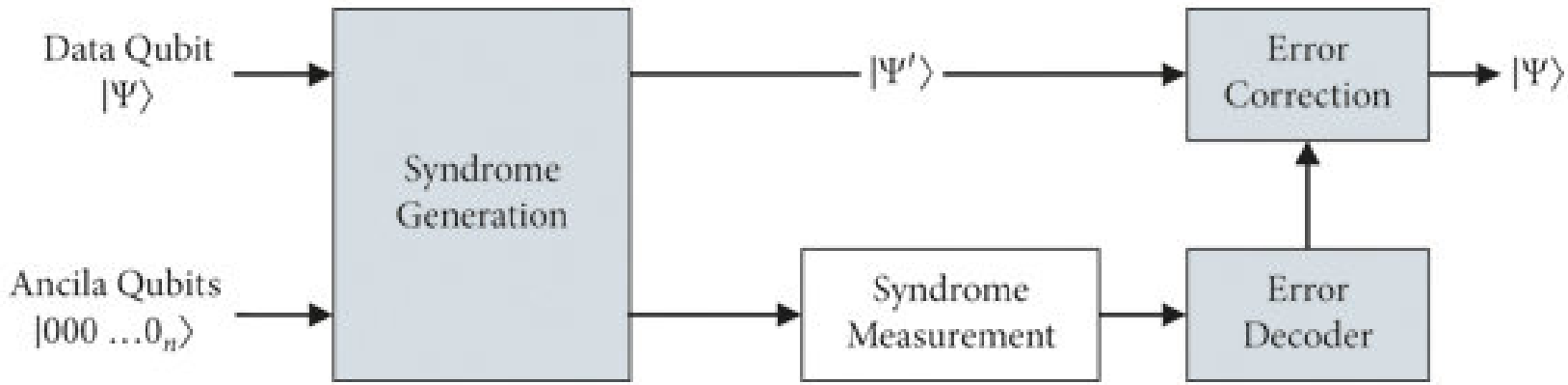
fluorescence detects leakage

1. Motivation

Detectability as a resource

Fault tolerance is a race between noise and decoding

- QEC spreads logical information across many physical qubits[1].
- Syndrome measurements constrain what errors could have happened.
- A **decoder** must infer a correction consistent with syndromes.



Baised noise

- Standard error model: depolarizing noise (X,Y,Z errors)
- Biased noise: one type of error (e.g., Z) is much more likely than others.

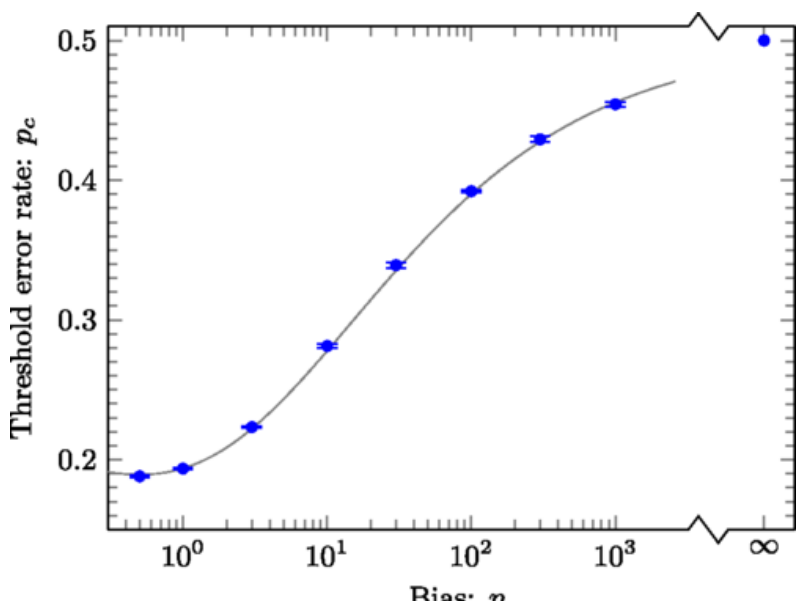


Figure 1: Threshold error rate vs bias[2]

- We find that the most biased condition have the highes error rate threshold about 50%.
- unbased condition: 18% threshold.(all perfect syndrome measurement)
- experimental condition: (errors+noisy operations) about 1% threshold. (surface code)[3]

Baised noise

Reason: The biased condition over one possible error reduced the space of possibilities for decoding syndrome measurements

Qubits:	1	2	3
Errors	X	X	X
	Y	Y	Y
	Z	Z	Z

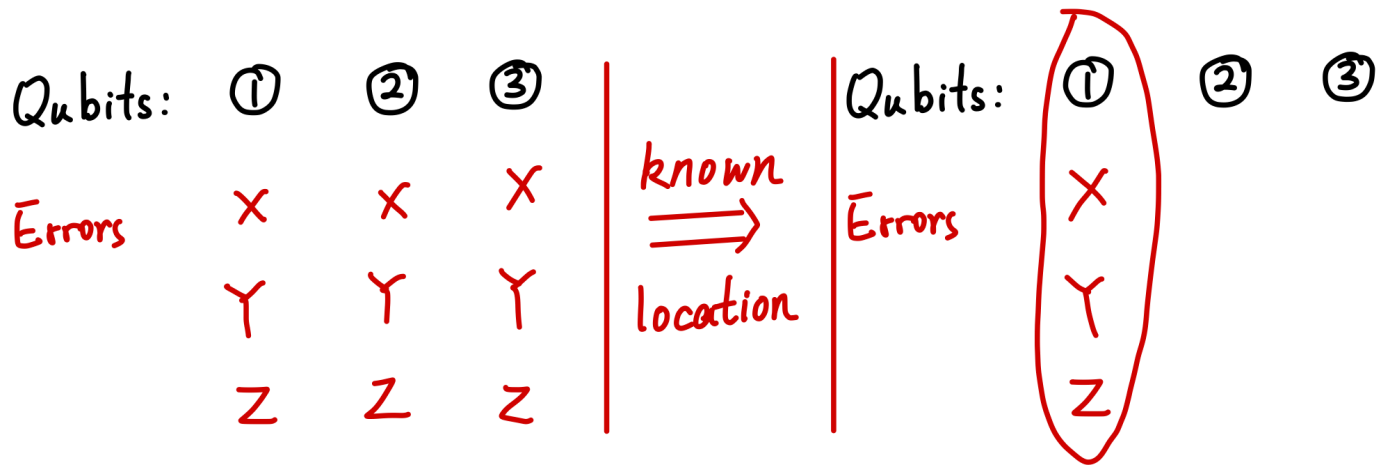
$\xRightarrow{\text{biased}}$

Z error

Qubits:	1	2	3
Errors			
	Z	Z	Z

Erasure errors

- An erasure error is a fault that occurs at a **known** location, but with an unknown Pauli type.
- Erasures are easier to correct: location of the error is avaliable to facilitate decoding.



Erasure errors can correct more errors

Key point erasure channel can detect more errors than depolarizing channel and correct **twice** as many errors

Depolarizing Channel $0 \leftrightarrow 1$	Erasure Channel $0 \to x$ $1 \to x$
0 errors 000 [detectable and correctable]	0 errors 000 [detectable and correctable]
1 errors 001, 010, 100 [detectable and correctable]	1 errors 00x, 0x0, x00 [detectable and correctable]
2 errors 011, 101, 110 [detectable]	2 errors 0xx, x0x, xx0 [detectable and correctable]
3 errors 111	3 errors xxx [detectable]
detect $(d-1)$ errors	detect d errors
correct $(d-1)/2$ errors	correct $(d-1)$ errors

Pauli errors vs erasures: what changes?

Same physics, different classical information

Unknown-location Pauli fault A qubit suffers $X/Z/Y$ (or 2-qubit Pauli) at an unknown site.

- Decoder must solve an inference problem: **which** sites failed?
- In surface-code decoding, ambiguity creates logical failures when many competing explanations exist.

Known-location erasure A fault occurred at a **known** site (location known; Pauli type may be unknown).

- Decoder treats erased sites as "missing / unreliable".
- Less ambiguity $\longrightarrow$ higher correctable rate at fixed code distance.

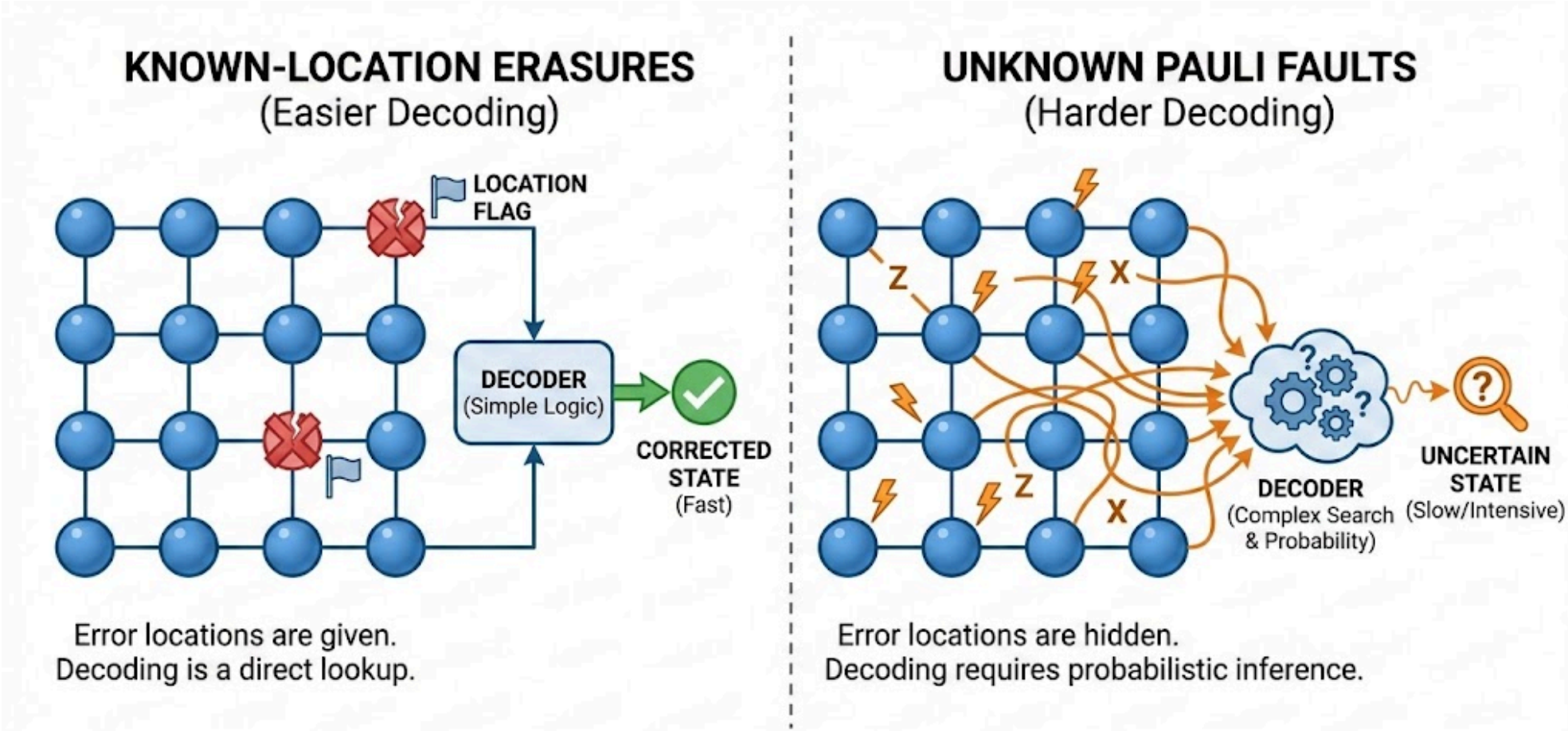


Figure 1: **Decoding Complexity Comparison.** Known-location erasures provide crucial side information, dramatically simplifying the correction process compared to blind Pauli faults.

Surface code context

- Surface code: 2D local checks; high practical threshold under Pauli noise (proposal: 1%).
- But "barely below threshold" is not enough—reasonable overhead needs p well below threshold.

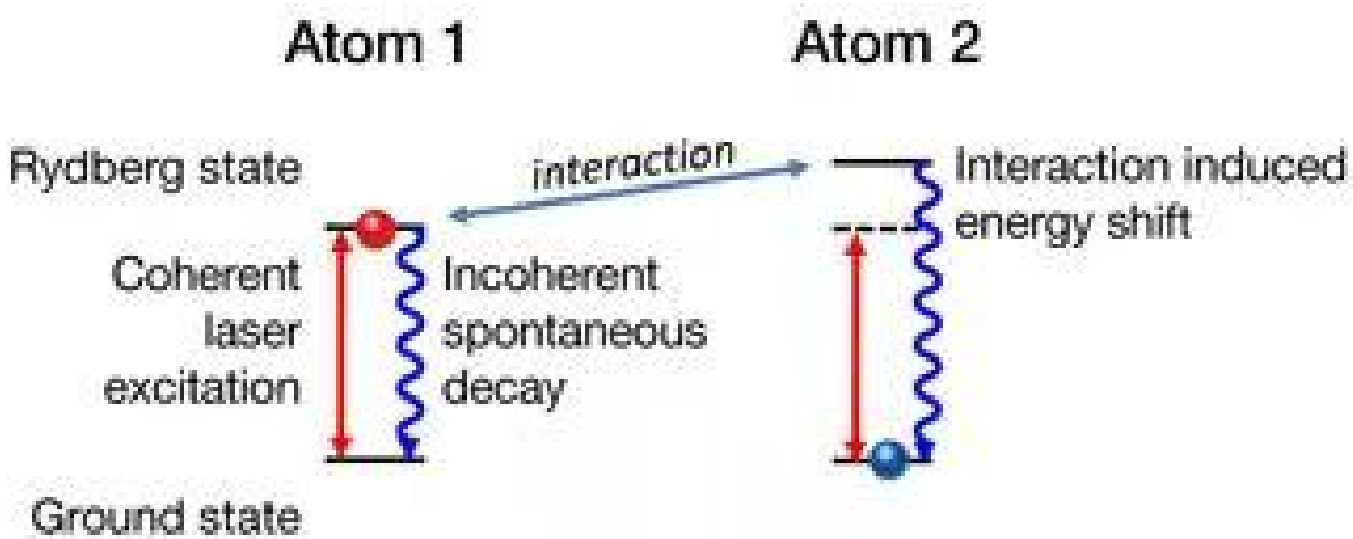
Motivating question Can the hardware **reveal** the location of its dominant errors, so the decoder's job becomes easier?

2. Physical layer

Metastable encoding makes valid qubits dark

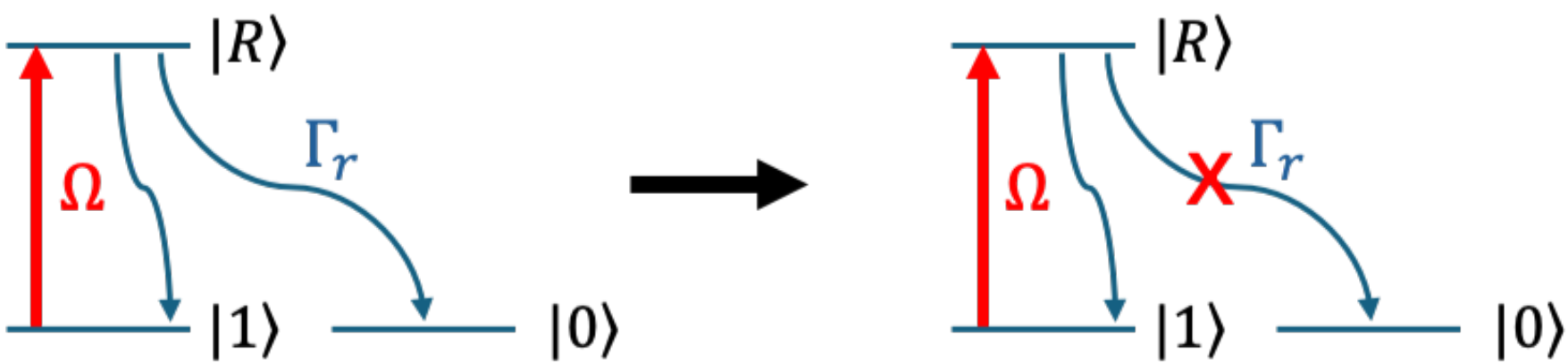
Error source in neutral atom platform

- Ground state atoms: long-lived, low error rates, but hard to entangle.
- Rydberg states: strong interactions enable fast gates, but short-lived and error-prone.
- Dominant gate fault: Rydberg decay during the gate.[4]

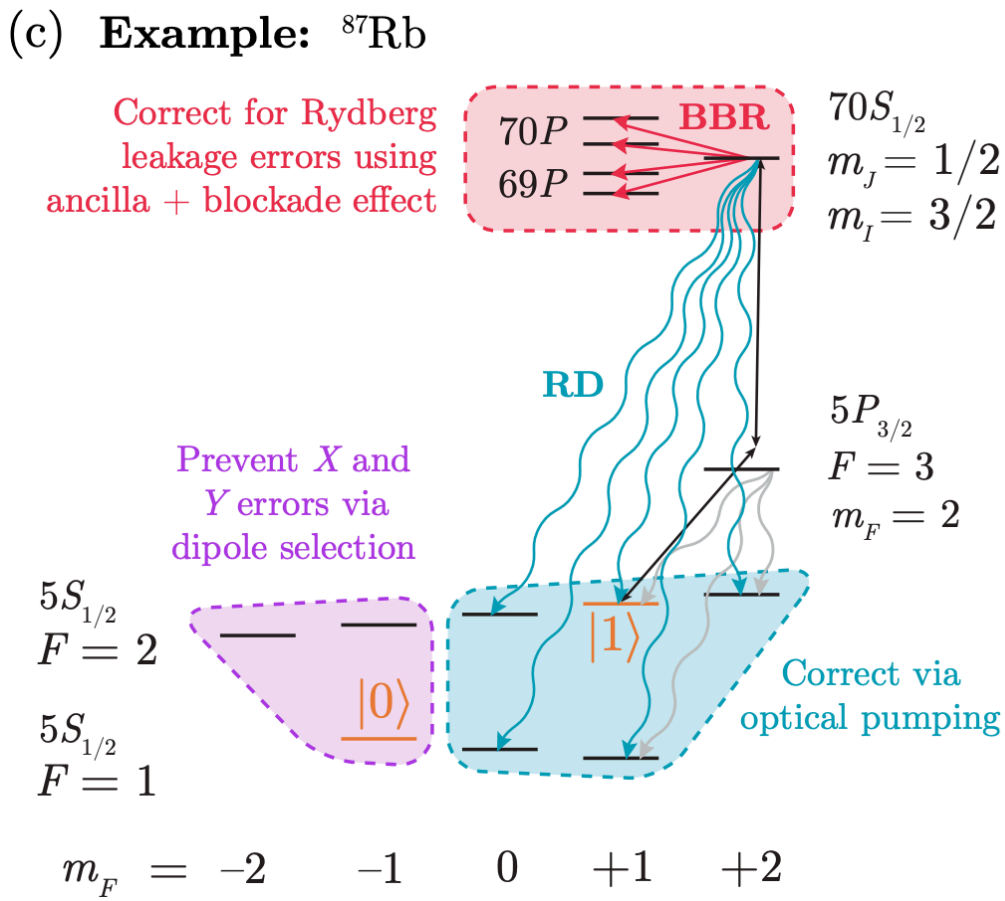


Rydberg decay channels for Rb atoms

- Encode qubits in a long-lived ground state of atoms Rb and Cs. Dominant gate fault (Rydberg decay) typically lands in the ground state manifold, becoming undetectable Pauli errors.

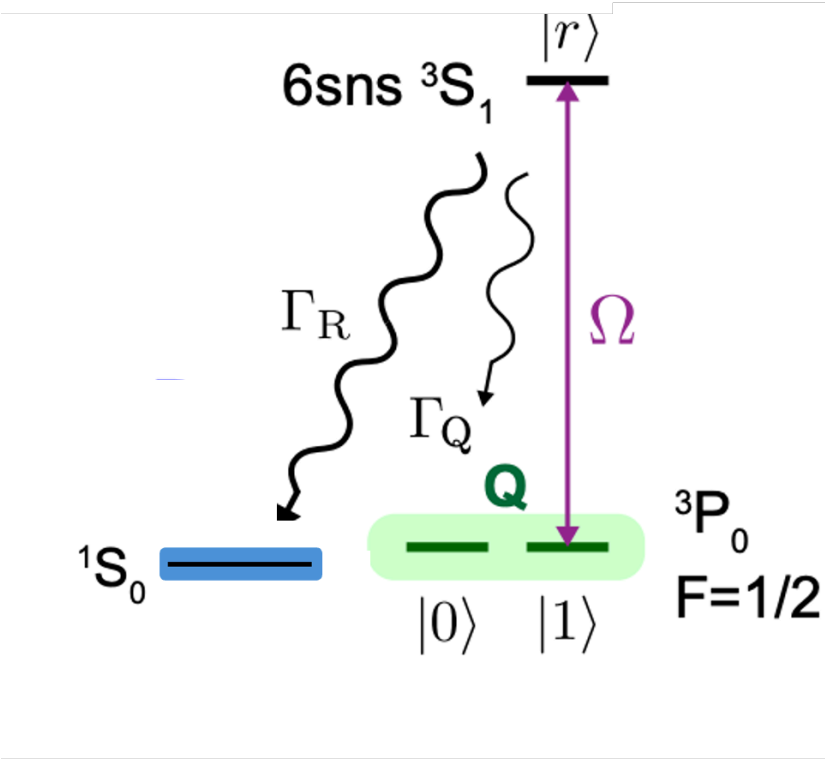


- Dipole selection + optical pumping for biased Z-error[5]



Erasure conversion for Ybs atoms

- What if we convert decay error all to erasure errors?

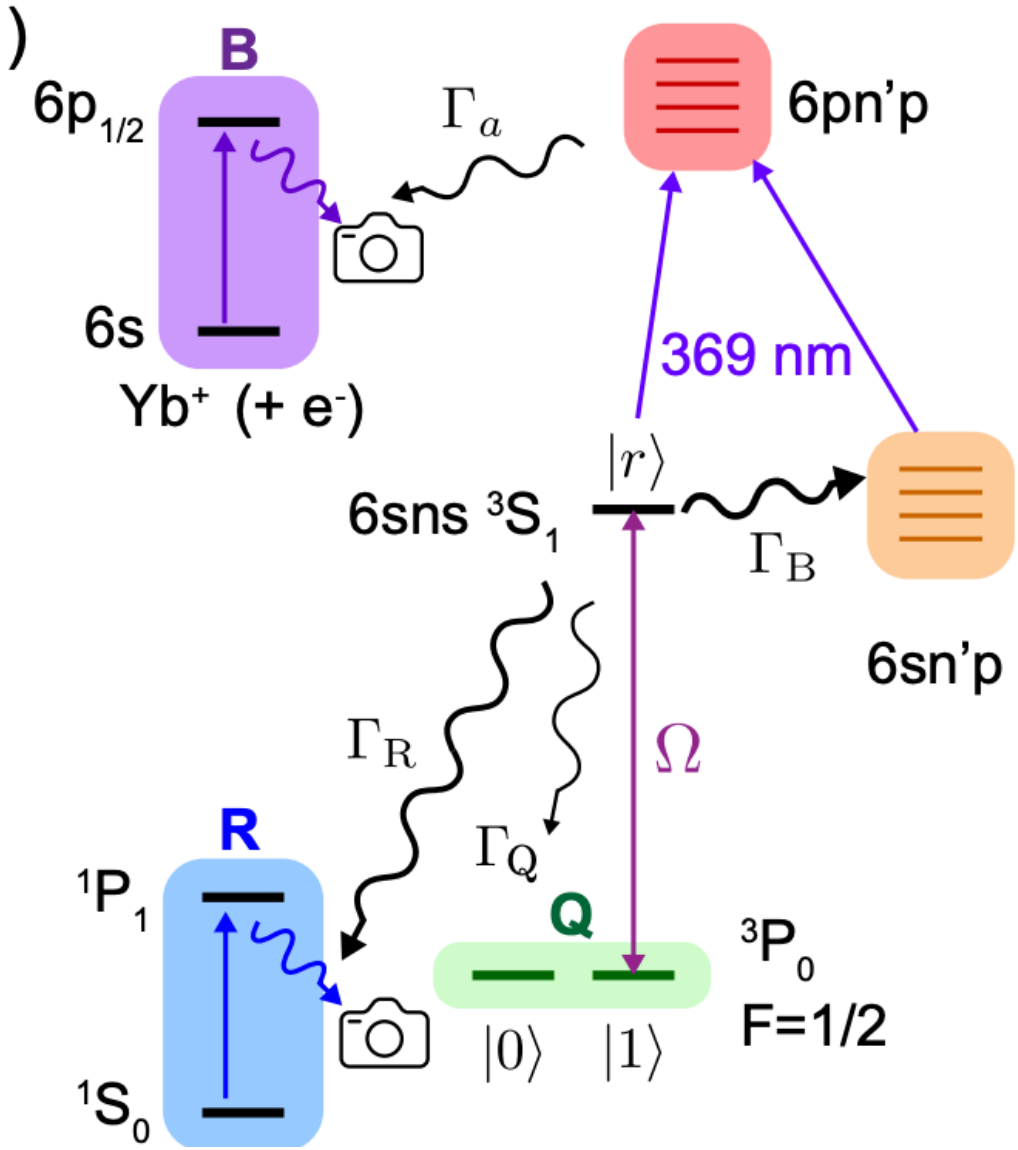


- Yb atoms, metastable states with long coherence times.
- Detect the population in the ground state manifold via fluorescence without discturbing the qubit manifold.

Condition for erasure conversion Absolute seperation of Computational subspace Q and dominant error subspace R (leakage); **detectable** signals without influencing Q.

Why 171-Yb enables erasure conversion

- **Metastable qubit storage:** encode in a long-lived metastable manifold $.^3P_0$ with a long lifetime (20 s).
- **Errors → erasures:** The dominate error mechanism (Rydberg decay) typically lands in the true ground state $.^1S_0$(detectable).
- **Non-destructive detection:** $.^1S_0$ is optically distinct from $.^3P_0$, so probing a cycling transition (e.g., $.^1S_0 \leftrightarrow .^1P_1$) makes leaked atoms sfluoresce (bright) while qubits in $.^3P_0$ remain dark.


)
B
6p1/2
6s
Yb+ (+ e-)
Γa
6pn'p
369 nm
|r⟩
6sns 3S1
ΓB
6sn'p
ΓR
ΓQ
Ω
R
1P1
1S0
Q
|0⟩
|1⟩
3P0
F=1/2

Question: what fraction of error become erasures?

Branching summary (proposal) 98% of relevant gate faults can be converted to erasures.

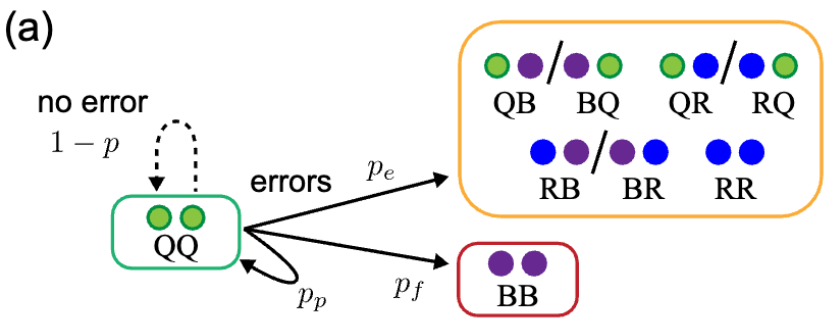
Parameterizing conversion

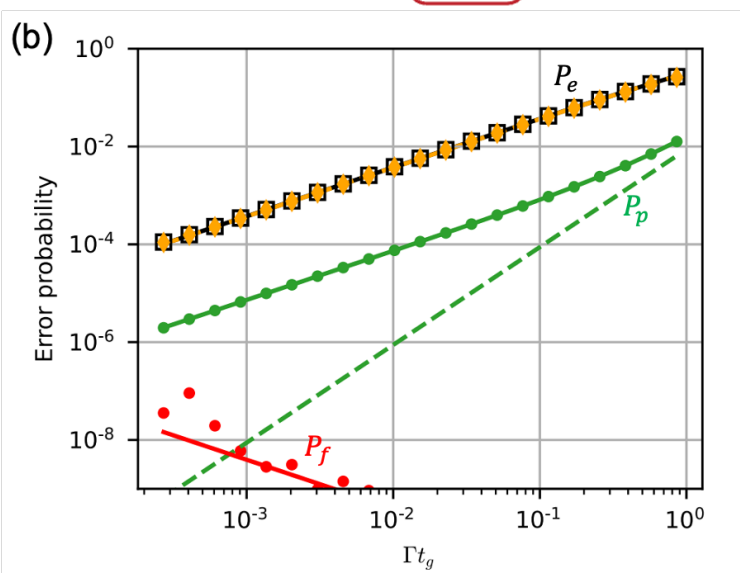
Let p_e = detected errors (erasures),

p_p = undetected errors (Pauli-like in Q-space).

Definition $R_e = \frac{p_e}{p_e + p_p}$

Possible error sources





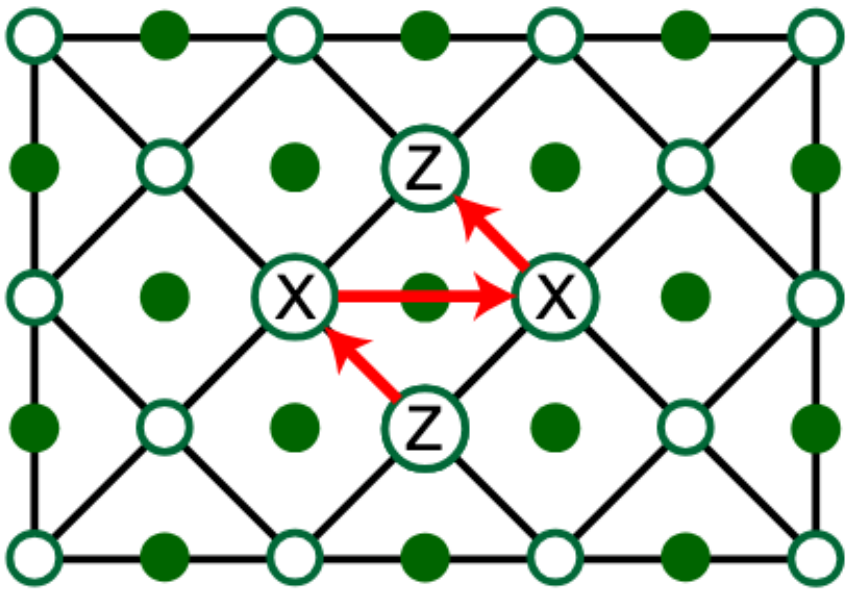
Three outcomes in the proposal

- Q: metastable qubit manifold (dark to detection light)
- R: ground manifold (bright under fluorescence)
- B: Rydberg state (autoionization and convert to Yb^+ and detect ion via fluorescence)

3. Protocol

Measure leakage after each gate layer

XZZX surface code



XZZX surface code

Parameterizing conversion

Assume errors only in two-qubit gates:

Pauli error with probability p_p
Erasure error with probability p_e.

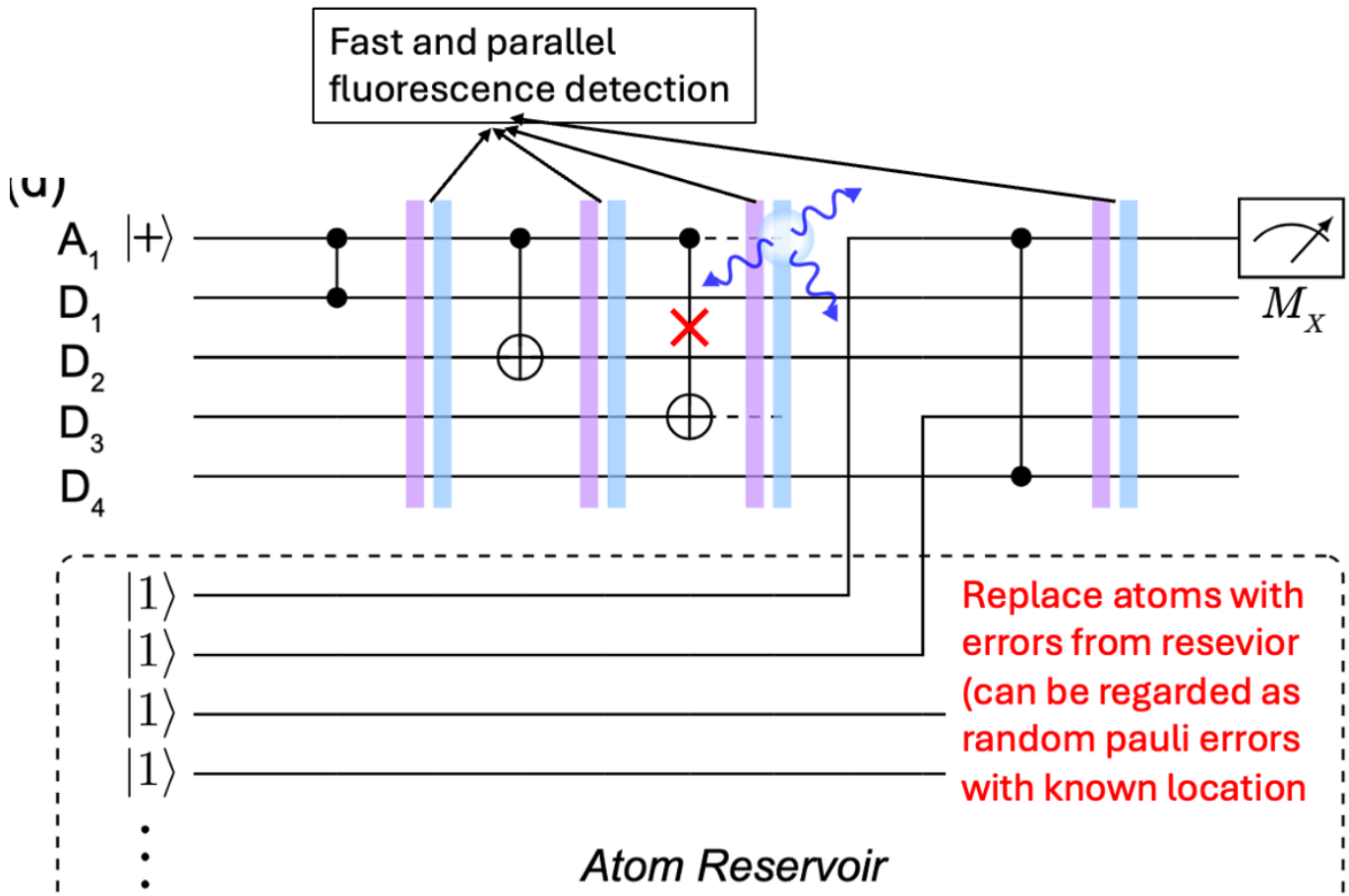
Erasure fraction $R_e = \frac{p_e}{p_e + p_p}$

XZZX Surface code

Key idea XZZX is a robust, simulation-friendly baseline for mixed noise + erasure, and pairs naturally with fast decoders.

1. **Robust across noise types:** a “safe baseline” under mixed/unknown noise.
2. XZZX integrates cleanly with a fast circuit-level decoder.
3. XZZX is well-suited when the Pauli type is uncertain (X/Z mixture).

Circuit-level integration



How is an erasure event represented in the circuit simulation?

Erasure = "location known, Pauli type unknown."

State model used in the simulator The erased qubit(s) are treated as being **replaced by a fresh qubit** in a maximally mixed state $\longrightarrow$ equivalently, a **uniformly random Pauli** is applied (type unknown), but the location is known.

4. QEC evidence

Threshold and scaling benefits

Main quantitative result: threshold jump

If we can convert most gate errors into erasures, how much did threshold improve in surface-code simulations?

Baseline (Pauli)

0.937%

surface-code threshold in proposal

With erasure conversion

4.15%

$R_e \approx 0.98$

Claimed implication Gate error rates in the 0.5–1% range move from “borderline” to “comfortably below threshold”.

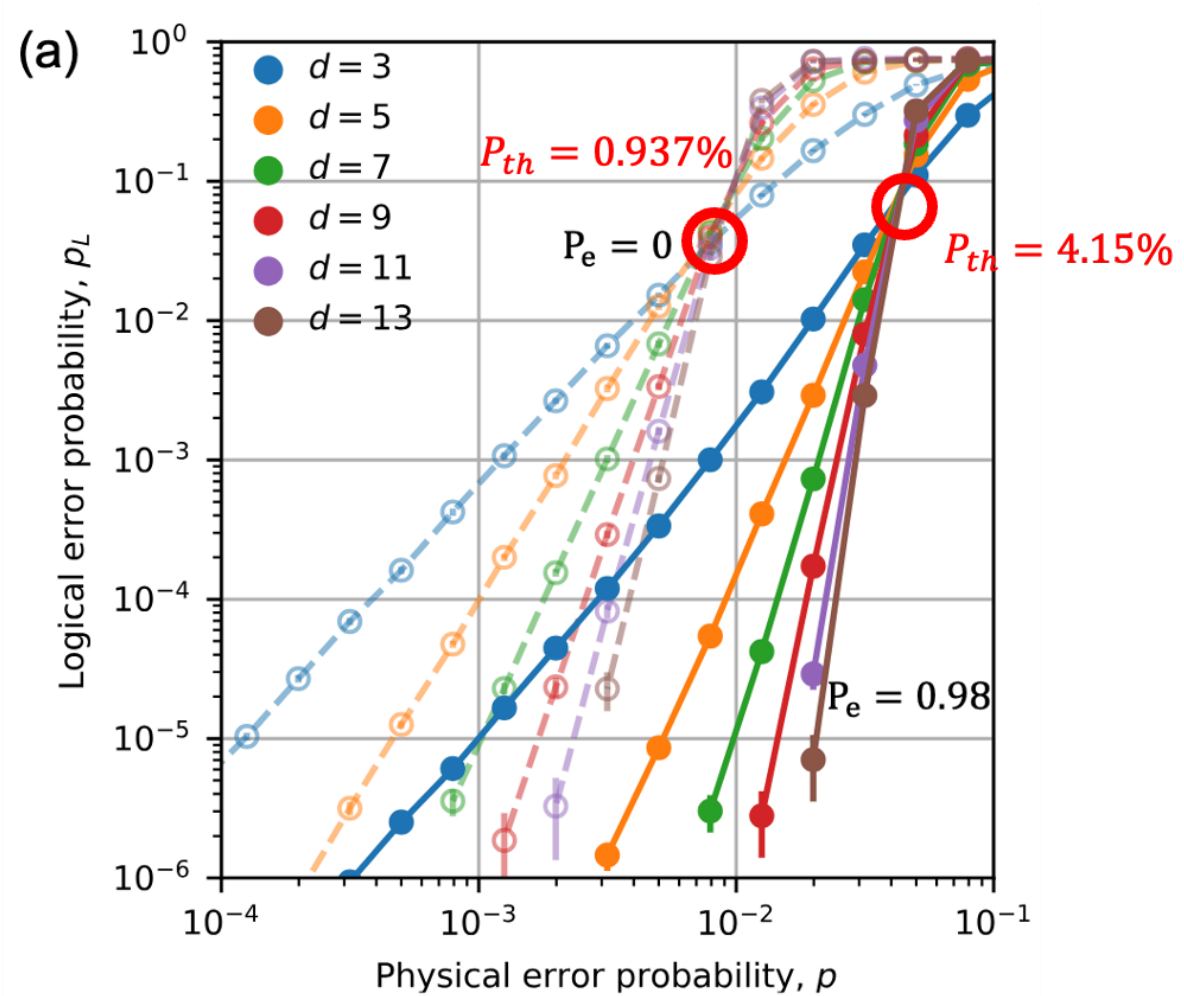
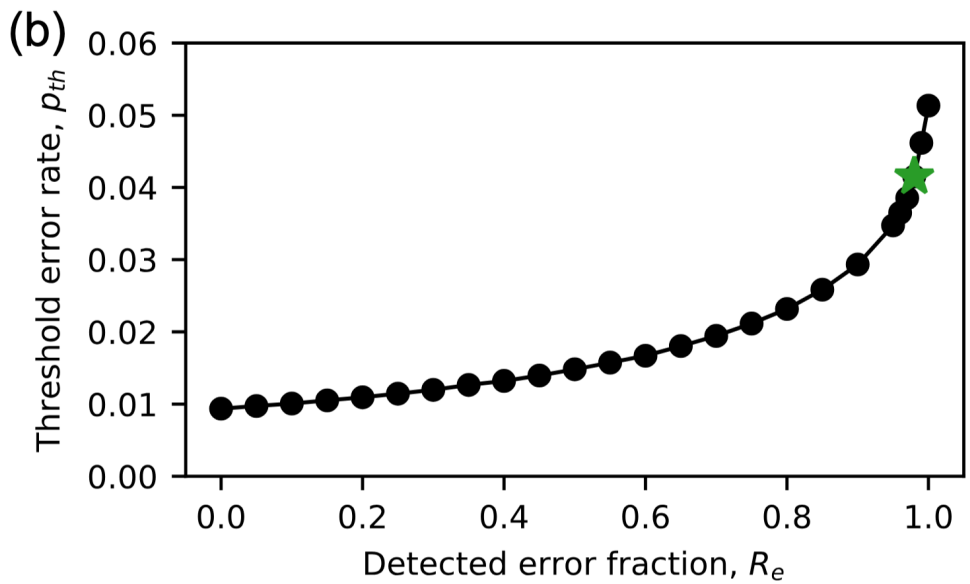


Figure 2: Numerical simulations using union find decoder.

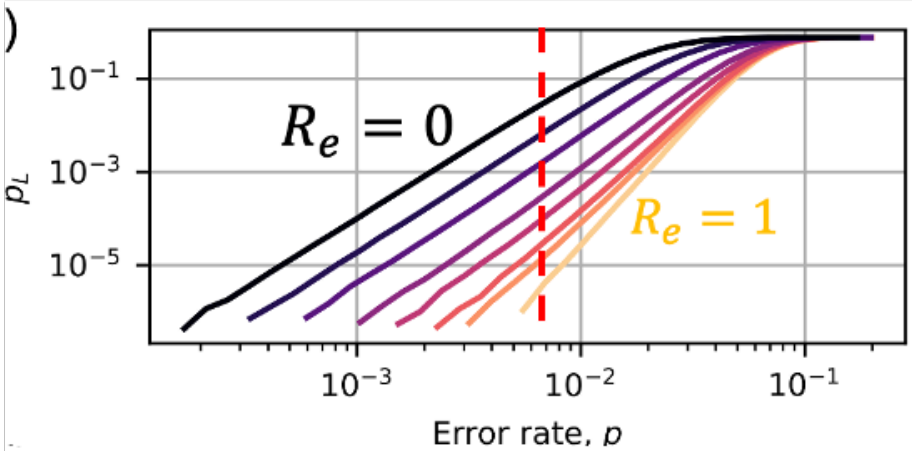
Interpretation Performance improves as R_e increases because the residual ambiguous component shrinks.

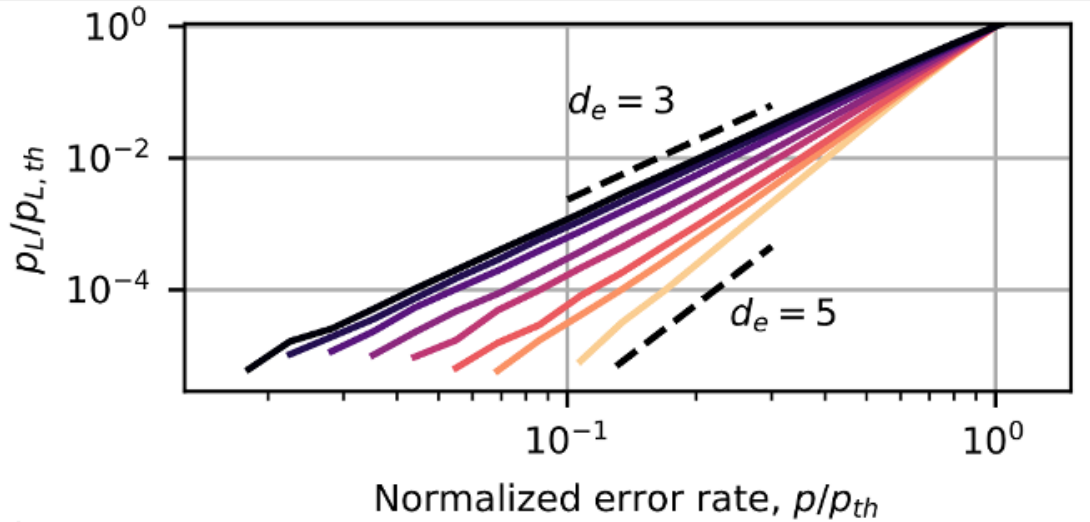


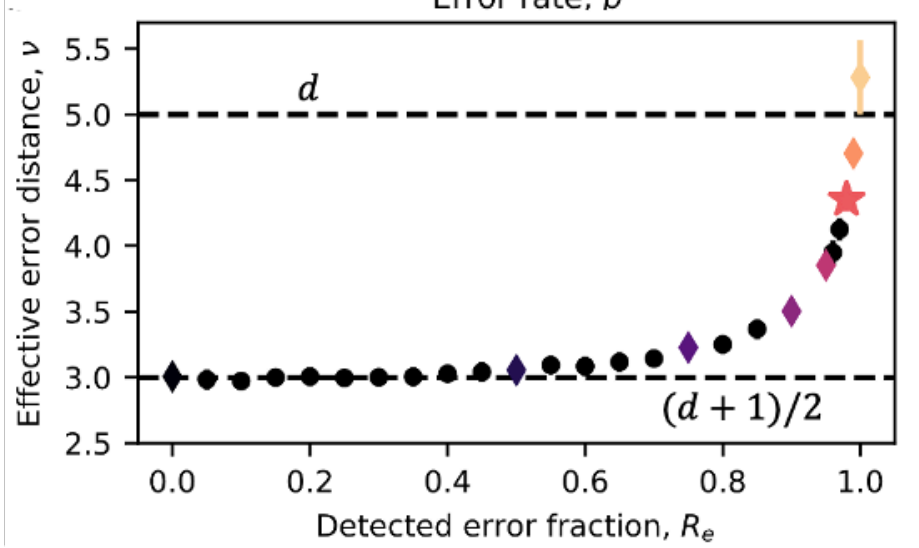
Why scaling improves: effective distance is preserved when locations are known

- For unknown-location Pauli faults, the code corrects up to $\frac{d-1}{2}$ faults (heuristic).
- For erasures (known locations), the code can tolerate up to $d-1$ erasures (heuristic).

Practical takeaway At fixed device size, conversion can yield lower logical error by steepening the logical-vs-physical scaling.







$$p_L \propto \mathrm{A} * \mathrm{p}^{\nu}$$

Pauli error → $\nu_{\mathrm{p}} = (d+1)/2$

Erasure error → $\nu_e = d$

Conclusion

Claim, requirements, next steps

Take-home messages

- **Claim:** Converting dominant gate faults into **flagged erasures** simplifies decoding and can substantially raise surface-code thresholds (as quoted earlier in the deck).
- **Requirement:** The benefit depends on **fast, high-fidelity leakage/loss detection** with low back-action and low latency (to keep the erasure fraction high).

Bottom line Erasure information acts like a hardware-side channel that can trade detection overhead for much larger fault-tolerance margins.

References

Bibliography

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